

ON THE LARGEST PRIME FACTOR OF QUADRATIC POLYNOMIALS

RUNBO LI

ABSTRACT. Let x denote a sufficiently large integer. We show that the recent result of Grimmelt and Merikoski actually implies that the largest prime factor of $n^2 + 1$ is greater than $x^{1.317}$ infinitely often. As an application, we give a new upper bound for the number of integers $n \leq x$ which $n^2 + 1$ has a primitive divisor.

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1. INTRODUCTION

Let x, n denote sufficiently large integers, p denote a prime number, P_r denote an integer with at most r prime factors counted with multiplicity, and let f be an irreducible polynomial with degree g . It's conjectured that there are infinitely many n such that $f(n)$ is prime. The simplest case is $g = 1$, which is the famous Dirichlet's theorem proved more than 100 years ago. However, for $g \geq 2$, this conjecture is still open.

For the second simplest case $g = 2$, there are several ways to attack this conjecture. One way is to relax the number of prime factors of $f(n)$, and the best result in this way is due to Iwaniec [10]. Building on the previous work of Richert [16], he showed that for any irreducible polynomial $f(n) = an^2 + bn + c$ with $a > 0$ and $c \equiv 1 \pmod{2}$, there are infinitely many x such that $f(x)$ has at most 2 prime factors.

Another possible way is to consider the largest prime factor of $f(n)$. Let $P^+(x)$ denote the largest prime factor of x , then we hope to show that the largest prime factor of $f(n)$ is greater than n^g for infinitely many integers n . For general polynomials, the best result is due to Tenenbaum [18], where he showed that for some $0 < t < 2 - \log 4$, the largest prime factor of $f(n)$ is greater than $n \exp((\log n)^t)$ for infinitely many integers n . However, it's rather difficult to prove the same thing holds for $n^{1+\varepsilon}$ even for a small ε .

For the special case $f(n) = n^2 + 1$, the progress is far more than the general case. In 1967, Hooley [9] first proved the largest prime factor of $n^2 + 1$ is greater than $n^{1.10014}$ for infinitely many integers n by using the Weil bound for Kloosterman sums. By applying their new bounds for multilinear forms of Kloosterman sums, Deshouillers and Iwaniec [2] showed in 1982 that the largest prime factor of $n^2 + 1$ is greater than $n^{1.202468}$ infinitely often. In 2020, de la Bretèche and Drappeau [1] improved the exponent to 1.2182 by making use of the result of Kim and Sarnak [11]. In 2023, Merikoski [13] proved a new bilinear estimate and used Harman's sieve to get the exponent 1.279. This is the first attempt of using Harman's sieve on this problem. In 2026, Pascadi [15] optimized the exponent to 1.3 by inserting his new arithmetic information. Recently, using a different approach to obtain arithmetical information, Grimmelt and Merikoski [7] got 1.312, which is the value that previously obtained by Merikoski [13] under the Selberg eigenvalue conjecture. In the present paper, we shall use the same sieve argument as in [7] and illustrate that this exponent can be further improved to 1.317.

Theorem 1.1. *Define*

$$\varrho_{a,h}(k) := \left| \left\{ \nu \in \mathbb{Z}/k\mathbb{Z} : a\nu^2 + h \equiv 0 \pmod{k} \right\} \right|.$$

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There exists some small $\varepsilon > 0$ such that the following holds for all $X > 1/\varepsilon$. Let $1 \leq h \leq X^{1+\varepsilon}$ be square-free and $1 \leq a \leq X^\varepsilon$ with $(a, h) = 1$. Suppose that for any $X^\varepsilon < Y < Z \leq X^2$ we have

$$\sum_{Y \leq p < Z} \frac{\log p}{p} \varrho_{a,h}(p) \leq (1 + \varepsilon) \log Z/Y + 1/\varepsilon.$$

Then there exists $n \in [X, 2X]$ such that the largest prime factor of $an^2 + h$ is greater than $X^{1.317+10^{-100}}$. For example, we know that the largest prime factor of $n^2 + 1$ is greater than $n^{1.317}$ infinitely often by taking $a = h = 1$ and

$$X^{1.317+10^{-100}} > (2X)^{1.317} \geq n^{1.317}.$$

As an application of our Theorem 1.1, we consider the polynomial $n^2 + 1$ with a primitive divisor.

Definition 1.2. Let (A_n) denote a sequence with integer terms. We say an integer $d > 1$ is a primitive divisor of A_n if $d \mid A_n$ and $(d, A_m) = 1$ for all non-zero terms A_m with $m < n$.

The next proposition is an easy result, and it can be proved in a same way of [[6], Lemma 4.1 and Proposition 4.2].

Proposition 1.3. ([5], Proposition 1.3). For all $n > 1$, the term $n^2 + 1$ has a primitive divisor if and only if $P^+(n^2 + 1) > 2n$. For all $n > 1$, if $n^2 + 1$ has a primitive divisor then that primitive divisor is a prime and it is unique.

Contrary to the previous works on the lower bounds for the largest prime factor, a result due to Schinzel [17] showed that for any $\varepsilon > 0$, the largest prime factor of $n^2 + 1$ is less than n^ε infinitely often. In fact, from his result we can easily get the following.

Theorem 1.4. ([5], Theorem 1.2). The polynomial $n^2 + 1$ does not have a primitive divisor for infinitely many terms.

We are interested in finding good upper and lower bound for the number of terms $n^2 + 1$ with a primitive divisor. We define

$$\rho(x) := |\{n \leq x : n^2 + 1 \text{ has a primitive divisor}\}|.$$

Then we have the following simple upper bound

$$\rho(x) < x - \frac{Cx}{\log x} \tag{1}$$

for some constant $C > 0$. In [6] the following stronger result is mentioned.

$$\rho(x) < x - \frac{x \log \log x}{\log x}. \tag{2}$$

In [5], Everest and Harman first proved a lower bound with positive density and a better upper bound for $\rho(x)$. More precisely, they got the following bounds:

Theorem 1.5. ([5], Theorem 1.4). We have

$$0.5324x < \rho(x) < 0.905x.$$

They also conjectured the asymptotic $\rho(x) \sim (\log 2)x$ in their paper. In 2024, Harman [8] used Merikoski's work on the largest prime factor of $n^2 + 1$ and sharpened the upper and lower bounds for $\rho(x)$.

Theorem 1.6. ([8], Theorem 5.5). We have

$$0.5377x < \rho(x) < 0.86x.$$

In the same year, Li [12] further improved the upper bound for $\rho(x)$ by using Pascadi's work. We remark that his calculation actually leads to an upper bound $0.85x$ instead of $0.8437x$ stated in his preprint.

Theorem 1.7. ([12], Theorem 1.6). We have

$$\rho(x) < 0.85x.$$

Mine [14] got a better lower bound for $\rho(x)$ in 2025.

Theorem 1.8. ([14], Corollary 1.5]). We have

$$\rho(x) > 0.543x.$$

In the present paper, we use the same sieve argument as in [12] and a recent result of Grimmelt and Merikoski to give a better upper bound for $\rho(x)$.

Theorem 1.9. We have

$$\rho(x) < 0.839x.$$

2. MERIKOSKI'S SIEVE DECOMPOSITIONS

Let ε denote a sufficiently small positive number and P_x denote the largest prime factor of $\prod_{x \leq n \leq 2x} (n^2 + 1)$. In this section we briefly introduce Grimmelt and Merikoski's work on finding a lower bound for P_x . Let $b = b(t)$ denote a non-negative smooth function supported on $[1, 4]$ and its derivatives satisfy $b^{(j)}(t) \ll 1$ for all $j \geq 0$. We define

$$|\mathcal{A}_d| := \sum_{n^2+1 \equiv 0 \pmod{d}} b\left(\frac{n}{x}\right) \quad \text{and} \quad X := \int b\left(\frac{t}{x}\right) dt = x \int b.$$

Then by the method of Chebyshev–Hooley and the discussions in [13], we only need to find an upper bound for

$$S(x) := \sum_{x < p \leq P_x} |\mathcal{A}_p| \log p = X \log x + O(x) \quad (3)$$

with a constant less than 1 to get a lower bound for P_x . By a smooth dyadic partition we have

$$S(x) = \sum_{\substack{x \leq P_j \leq P_x \\ P_j = 2^j x}} S(x, P_j) + O(x), \quad (4)$$

where

$$S(x, P_j) = \sum_{P_j \leq p \leq 4P_j} \psi_j\left(\frac{p}{P_j}\right) |\mathcal{A}_p| \log p \quad (5)$$

for a smooth function $\psi_j = \psi_j(t)$ supported on $[1, 4]$ satisfying $\psi_j^{(l)}(t) \ll 1$ for all $l \geq 0$.

In [7], Grimmelt and Merikoski proved the following upper bound for $S(x)$ with $P_x = x^{1.312}$ by using Harman's sieve method together with their new arithmetic information.

Lemma 2.1. (See [13], [7]). We have

$$\begin{aligned} \sum_{\substack{x \leq P_j \leq x^{1.312} \\ P_j = 2^j x}} S(x, P_j) &\leq (G_0 + G_1 + G_2 + G_3 + G_4 + G_5 - G_6 + G_7) X \log x \\ &< 0.998X \log x, \end{aligned}$$

where

$$\begin{aligned} G_0 &= \int_1^{\frac{7}{6}} 1 d\alpha = \frac{1}{6}, \\ G_1 &= \int_1^{\frac{17}{16}} \int_{\sigma(\alpha)}^{\alpha-2\sigma(\alpha)} \alpha \frac{\omega\left(\frac{\alpha-\beta}{\beta}\right)}{\beta^2} d\beta d\alpha + \int_1^{\frac{17}{16}} \int_{\xi(\alpha)}^{\frac{\alpha}{2}} \alpha \frac{\omega\left(\frac{\alpha-\beta}{\beta}\right)}{\beta^2} d\beta d\alpha < 0.0287, \\ G_2 &= \int_{\frac{17}{16}}^{\frac{8}{7}} \int_{\sigma(\alpha)}^{\frac{\alpha}{2}} \alpha \frac{\omega\left(\frac{\alpha-\beta}{\beta}\right)}{\beta^2} d\beta d\alpha < 0.08622, \\ G_3 &= \int_{\frac{8}{7}}^{\frac{7}{6}} \int_{\sigma(\alpha)}^{\frac{\alpha}{2}} \alpha \frac{\omega\left(\frac{\alpha-\beta}{\beta}\right)}{\beta^2} d\beta d\alpha < 0.03107, \end{aligned}$$

$$\begin{aligned}
G_4 &= \int_{\frac{8}{7}}^{\frac{7}{6}} \int_{\sigma(\alpha)-\alpha+1}^{\alpha-1} \int_{\sigma(\alpha)-\alpha+1}^{\beta_1} \int_{\sigma(\alpha)-\alpha+1}^{\beta_2} f_4(\alpha, \beta_1, \beta_2, \beta_3) \alpha \frac{\omega\left(\frac{\alpha-\beta_1-\beta_2-\beta_3}{\beta_3}\right)}{\beta_1\beta_2\beta_3^2} d\beta_3 d\beta_2 d\beta_1 d\alpha < 0.00011, \\
G_5 &= 4 \int_{\frac{7}{6}}^{\frac{5}{4}} \alpha d\alpha = \frac{29}{72}, \\
G_6 &= \int_{\frac{7}{6}}^{\frac{5}{4}} \int_{\alpha-1}^{\sigma(\alpha)} \alpha \frac{\omega\left(\frac{\alpha-\beta}{\beta}\right)}{\beta^2} d\beta d\alpha > 0.035631, \\
G_7 &= 4 \int_{\frac{5}{4}}^{1.312} \alpha d\alpha < 0.31769,
\end{aligned}$$

where

$$\sigma(\alpha) := \frac{2-\alpha}{3}, \quad \xi(\alpha) = \frac{3}{2} - \alpha, \quad (6)$$

f_4 denotes the characteristic function of the set

$$\{\beta_1 + \beta_2, \beta_1 + \beta_3, \beta_2 + \beta_3, \beta_1 + \beta_2 + \beta_3 \notin [\alpha - 1, \sigma(\alpha)]\},$$

and $\omega(u)$ denotes the Buchstab function determined by the following differential-difference equation

$$\begin{cases} \omega(u) = \frac{1}{u}, & 1 \leq u \leq 2, \\ (u\omega(u))' = \omega(u-1), & u \geq 2. \end{cases}$$

However, their bounds for those integrals are not very accurate. Using Mathematica 14 (see the ancillary files for arXiv:2406.07575v2 for the code), we can get the following better bounds. These bounds are verified by the work of Drappeau and Mounier [4] [3] using **LattE**. For a comparison, the bounds in brackets are obtained by Merikoski [13]. We remark that for G_6 the new lower bound gives a 67% improvement over the bound mentioned in [13].

Lemma 2.2. For G_i ($0 \leq i \leq 6$) defined in Lemma 2.1, we have

$$\begin{aligned}
G_0 &= \frac{1}{6}, \quad 0.02861 < G_1 < 0.028611(0.0287), \quad 0.08606 < G_2 < 0.086062(0.08622), \\
0.03099 &< G_3 < 0.030992(0.03107), \quad 0 < G_4 < 0.0001(0.00011), \\
G_5 &= \frac{29}{72}, \quad 0.0599 > G_6 > 0.059841(0.035631).
\end{aligned}$$

Moreover, with these new bounds for G_i ($0 \leq i \leq 6$) we have

$$G_0 + G_1 + G_2 + G_3 + G_4 + G_5 - G_6 + 4 \int_{\frac{5}{4}}^{1.317+10^{-100}} \alpha d\alpha < 0.9999$$

and

$$G_0 + G_1 + G_2 + G_3 + G_4 + G_5 - G_6 + 4 \int_{\frac{5}{4}}^{1.3173} \alpha d\alpha > 1.0001.$$

By Lemma 2.2 and the same arguments as in [7], we complete the proof of Theorem 1.1.

3. PROOF OF THEOREM 1.9

By the discussions in [5] and [8], we wish to get an upper bound of the form

$$\sum_{1 \leq p x^{-\alpha} \leq e} |\mathcal{A}_p| \leq K(\alpha)(1 + o(1)) \frac{X}{\log x},$$

where $K(\alpha)$ is the sum of sieve theoretical functions related to the sieve decomposition on the problem of the largest prime factor of $n^2 + 1$. This requires us to find the maximum value for τ such that

$$\int_1^\tau \alpha K(\alpha) d\alpha < 1.$$

By Lemma 2.2 we know that $\tau < 1.3173$, and $K(\alpha)$ can be defined as the piecewise function in Section 2:

$$K(\alpha) := \mathbb{1}_{[1, \frac{7}{6})} \cdot \frac{1}{\alpha} + \mathbb{1}_{[1, \frac{17}{16})} \cdot \left(\int_{\sigma(\alpha)}^{\alpha-2\sigma(\alpha)} \frac{\omega\left(\frac{\alpha-\beta}{\beta}\right)}{\beta^2} d\beta + \int_{\xi(\alpha)}^{\frac{\alpha}{2}} \frac{\omega\left(\frac{\alpha-\beta}{\beta}\right)}{\beta^2} d\beta \right) + \cdots + 4 \cdot \mathbb{1}_{[\frac{5}{4}, \frac{3}{2})}.$$

Combining this with the bound proved in [5], we have

$$\begin{aligned} \rho(x) &< (1 + o(1))x \int_1^{1.3173} K(\alpha) d\alpha \\ &\leq (G'_0 + G'_1 + G'_2 + G'_3 + G'_4 + G'_5 - G'_6 + G'_7) x, \end{aligned} \quad (7)$$

where

$$\begin{aligned} G'_0 &= \int_1^{\frac{7}{6}} \frac{1}{\alpha} d\alpha = \log \frac{7}{6} < 0.154151, \\ G'_1 &= \int_1^{\frac{17}{16}} \int_{\sigma(\alpha)}^{\alpha-2\sigma(\alpha)} \frac{\omega\left(\frac{\alpha-\beta}{\beta}\right)}{\beta^2} d\beta d\alpha + \int_1^{\frac{17}{16}} \int_{\xi(\alpha)}^{\frac{\alpha}{2}} \frac{\omega\left(\frac{\alpha-\beta}{\beta}\right)}{\beta^2} d\beta d\alpha < 0.027475, \\ G'_2 &= \int_{\frac{17}{16}}^{\frac{8}{7}} \int_{\sigma(\alpha)}^{\frac{\alpha}{2}} \frac{\omega\left(\frac{\alpha-\beta}{\beta}\right)}{\beta^2} d\beta d\alpha < 0.077933, \\ G'_3 &= \int_{\frac{8}{7}}^{\frac{7}{6}} \int_{\sigma(\alpha)}^{\frac{\alpha}{2}} \frac{\omega\left(\frac{\alpha-\beta}{\beta}\right)}{\beta^2} d\beta d\alpha < 0.026835, \\ G'_4 &= \int_{\frac{8}{7}}^{\frac{7}{6}} \int_{\sigma(\alpha)-\alpha+1}^{\alpha-1} \int_{\sigma(\alpha)-\alpha+1}^{\beta_1} \int_{\sigma(\alpha)-\alpha+1}^{\beta_2} f_4(\alpha, \beta_1, \beta_2, \beta_3) \frac{\omega\left(\frac{\alpha-\beta_1-\beta_2-\beta_3}{\beta_3}\right)}{\beta_1\beta_2\beta_3^2} d\beta_3 d\beta_2 d\beta_1 d\alpha < 0.00009, \\ G'_5 &= 4 \int_{\frac{7}{6}}^{\frac{5}{4}} 1 d\alpha = \frac{1}{3}, \\ G'_6 &= \int_{\frac{7}{6}}^{\frac{5}{4}} \int_{\alpha-1}^{\sigma(\alpha)} \frac{\omega\left(\frac{\alpha-\beta}{\beta}\right)}{\beta^2} d\beta d\alpha > 0.05016, \\ G'_7 &= 4 \int_{\frac{5}{4}}^{1.3173} 1 d\alpha = 0.2692. \end{aligned}$$

By a simple calculation, the value of the right hand side of (7) is less than $0.839x$, and the proof of Theorem 1.9 is completed.

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INTERNATIONAL CURRICULUM CENTER, THE HIGH SCHOOL AFFILIATED TO RENMIN UNIVERSITY OF CHINA, BEIJING, CHINA
 Email address: `runbo.li.carey@gmail.com`